Statistics Physics

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Capítulo 1

Introduction

Statistical physics is the branch of physics that studies systems composed of a large number of constituents (or *degrees of freedom*). Its objective is to establish a connection between the essential properties of the constituents (e.g. atoms, molecules, particles) of a given system and its macroscopic or collective properties. Theoretically, one might determine the complete set of relevant properties of a system if positions and velocities of its constituents were completely known, that is if their dynamics and a model describing their mutual interactions were available. However, this approach would be immediately rejected for two main reasons: (1) describing the time evolution of, say, one mole (roughly 10^{23}) of particles is technically impossible and (2) the time needed to observe a system and measure its properties in a given configuration is much longer than the characteristic time over which that system changes on a microscopic scale due to the collisions between its constituents. In other words, by the time we measure position and velocity of each atom, molecule or particle, these positions and velocities have already changed many times and the system has explored a huge number of *microstates*. What one actually measures in experiments is an average of individual microstates. Wetherefore conclude that mapping the transition dynamics between two subsequent microstates is not only computationally prohibitive, but also uninteresting. Therefore, it makes more sense to seek a statistical approach. (Patti, 2025)

1.1. Microscopic and Macroscopic Perspectives

Statistical physics is a branch of Theoretical Physics that aims to provide an explanation for the macro scopic properties of matter (mechanical, thermal, electrical, magnetic, etc.) based on the knowledge of the microscopic properties of its constituents. It employs concepts of **probability** to establish the connection between the microscopic and macroscopic universe.

To understand the behaviour of a system at the **microscopic** level, we need an interpretative model that describes the particles and their interactions. This model can be based on either *Classical Mechanics* or *Quantum Mechanics*, depending on the nature of the system. In either case, we rely on the system's **Hamiltonian**, which represents its energy and governs its dynamics. One striking characteristic of the microscopic realm is the **vast** number of accessible microscopic states. The system's particles exhibit fluctuating and chaotic behaviour, with their positions and velocities **rapidly fluctuating** over time. This dynamic nature arises from various factors, such as thermal motion, quantum effects, and the intricate interactions among particles. As already noticed above, mathematically resolving the precise motion of particles at this scale is infeasible due to the **enormous** number of particles involved. Even if we had complete knowledge of their individual motions, it would not be practical for predicting the macroscopic behaviour of the system. This is because the macroscopic properties of the system emerge from the collective behaviour of an enormous number of particles, making it challenging to directly deduce macroscopic phenomena from microscopic dy namics. Furthermore, the experimental observation of the microscopic state of a system is **impossible in practice**. Even if it were hypothetically feasible, the sheer volume of data would be overwhelming. Handling NA number of particles is prohibitively demanding, even for the most powerful supercomputers available today. Hence, **statistical physics offers an alternative approach**. Instead of focusing on individual particle details, it seeks to describe and understand the behaviour of macroscopic systems through statistical analysis and probability. By examining the statistical properties of ensembles of particles, statistical physics allows us to derive macroscopic laws and make predictions about observable phenomena.

By contrast, when we examine a system from a **macroscopic** viewpoint, we focus on **global quantities** that describe the system as a whole. These quantities provide a comprehensive understanding of the system's macroscopic state. In this context, we typically consider only a handful of variables that capture essential aspects of the system's behaviour. For instance, variables like pressure (ρ), temperature (T), volume (V), internal energy (U), and entropy (S) are commonly used to characterise and analyse macroscopic systems. Unlike the microscopic level, where particles exhibit rapid and chaotic fluctuations in their positions and velocities, macroscopic properties tend to be more stable and consistent. For example, when we repeatedly measure the pressure of a gas in equilibrium, we expect to obtain the same result each time, indicating a predictable and non-fluctuating macroscopic behaviour. Importantly, macroscopic properties are directly accessible through experimental observations. We can measure and quantify these properties in real-world experiments, enabling us to gain insights into the behaviour and properties of macroscopic systems. It is essential to recognise that, when we perform measurements on macroscopic systems, the process takes time. Unlike the idealised concept of an instant measurement, obtaining an accurate measurement of a macroscopic property requires a finite interval of time, τ. During this time interval, the system may undergodynamic changes or fluctuations, which can arise from various sources, such as the inherent thermal motion of particles, external influences, or the system transitioning between different states. (Patti, 2025)

1.1.1. Subsección

Nota

En la realidad no existen rayos monocromáticos debido al principio de incertidumbre de Heisenberg, los rayos siempre tienen un ancho de banda $\Delta\nu$

Bibliografía

Patti, A. (2025). Lectures notes in Statistical Physics.